EPI 501: Principles of Epidemiology
Synthetic sample syllabus for PHACT testing. Not a real course.

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Course description
This course introduces the core methods of epidemiology, including measures of disease frequency, study design, bias, confounding, and causal inference. Students apply these methods to current public health problems using real surveillance data sets.

Learning objectives
By the end of the course, students will be able to:
1. Calculate and interpret measures of disease frequency and association.
2. Compare the strengths and limitations of observational and experimental study designs.
3. Identify sources of bias and confounding in published studies.
4. Analyze surveillance data to describe patterns of disease in populations.
5. Communicate epidemiologic findings to professional and lay audiences.

Weekly schedule
Week 1: History and scope of epidemiology
Week 2: Measures of disease frequency
Week 3: Measures of association
Week 4: Cross-sectional and ecological studies
Week 5: Cohort studies
Week 6: Case-control studies
Week 7: Randomized trials
Week 8: Bias
Week 9: Confounding and effect modification
Week 10: Screening and diagnostic testing
Week 11: Surveillance and outbreak investigation
Week 12: Causal inference
Week 13: Communicating findings
Week 14: Final project presentations

Assessments
Problem sets (5): 30%
Critical appraisal of a published study: 20%
Midterm exam: 20%
Final project: surveillance data analysis and policy brief: 30%

Course policies
Late work loses 10% per day. Collaboration on problem sets is allowed, but each student submits their own work. Academic integrity policies of the school apply to all assignments.